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TruthLens: A Hybrid AI Framework for Real-Time
Misinformation Detection via Multi-Source Scrapers
and Machine Learning Pipelines

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Abstract—The rapid growth of digital media and online news platforms has significantly increased the spread of misinforma-tion and fake news, posing serious risks to public trust and societal stability. Traditional fact-checking methods are no longer sufficient due to the high volume and speed of information generation, creating a need for automated and scalable detection systems.

This paper presents TruthLens, a machine learning-based fake news detection system designed for real-time analysis of news content. The system integrates natural language processing techniques with a full-stack architecture to classify news as real or fake. It supports both direct text input and URL-based article analysis, enabling dynamic extraction and evaluation of live content. Feature extraction is performed using TF-IDF, followed by classification using a supervised machine learning model trained on a balanced dataset of real and fake news articles. The backend is implemented using FastAPI for efficient model inference, while a user-friendly React-based frontend enables interactive analysis. Supabase is used for authentication, data storage, and feedback management. Experimental results demonstrate reliable classification performance of 92% accuracy, highlighting the effectiveness of the proposed approach. The system's scalable and deployment-ready design makes it suitable for real-world applications in combating misinformation.

Future work includes enhancing the model using deep learn-ing techniques, supporting multilingual data, and incorporating multimodal analysis for improved accuracy.

Index Terms—Fake News Detection, Natural Language Pro-cessing, FastAPI, TF-IDF Vectorization, Full-Stack AI, Digital

Trust, Web Scraping, Passive-Aggressive Classifier.

I. INTRODUCTION

The rapid growth of digital media and online news platforms has significantly transformed the way information is disseminated and consumed. While this evolution has improved accessibility and speed, it has also led to the widespread proliferation of misinformation and fake news. Fake news refers to deliberately fabricated or misleading information presented as legitimate news, often created to influence public opinion, generate revenue, or create social disruption. The increasing reliance on social media platforms as primary news sources has further amplified the reach and impact of such misinformation, posing serious threats to societal trust, political stability, and public safety.

Traditional methods of detecting fake news, such as manual fact-checking and editorial review, are no longer sufficient due to the sheer volume and velocity of information being generated daily. As a result, there is a growing need for automated, scalable, and intelligent systems capable of identifying misinformation in real time. Recent advancements in machine learning and natural language processing (NLP) have enabled the development of models that can analyze textual patterns and classify news content with reasonable accuracy.

Despite these advancements, existing solutions exhibit several limitations. Many current systems rely heavily on static datasets and lack the ability to generalize effectively to real-world, dynamic content. Furthermore, a significant number of approaches focus solely on textual input and do not support real-time analysis of live news articles through URLs. Additionally, existing implementations often lack integration with user-centric features such as authentication, feedback mechanisms, and scalable deployment architectures, limiting their practical applicability.

To address these challenges, this paper proposes TruthLens, a full-stack fake news detection system that integrates machine learning with modern web technologies to provide real-time and user-friendly misinformation analysis. The system accepts both raw text and URL-based inputs, enabling dynamic extraction and classification of news content. A pre-trained machine learning model, combined with TF-IDF feature extraction, is deployed through a FastAPI backend to perform efficient inference. The system further incorporates a React-based frontend for interactive user engagement and Supabase for authentication and data management.

The key contributions of this work are summarized as follows:

- Design and development of an end-to-end fake news detection system integrating frontend, backend, and machine learning components.
- Implementation of real-time news analysis supporting both direct text input and URL-based content extraction.
- Deployment of a scalable backend architecture using FastAPI for efficient model inference.
- Integration of user authentication and data storage using

Supabase to enhance usability and personalization.
• Provision of a feedback mechanism to enable continuous improvement of the system.
• Mathematical formulation of core NLP components including TF-IDF and classification boundaries.
The proposed system aims to bridge the gap between theoretical machine learning models and practical, deployable applications, contributing toward more effective and accessible solutions for combating misinformation in the digital age.
II. LITERATURE SURVEY
The problem of fake news detection has been extensively studied using classical machine learning approaches, particularly focusing on textual feature extraction and efficient classification algorithms.
Ahmed et al. [1] explored linguistic stylometry techniques for fake news detection by utilizing n-gram features combined with traditional classifiers such as Logistic Regression and Linear Support Vector Machines (SVM). Their study demonstrated that simple frequency-based textual features can achieve accuracy levels exceeding 90%, highlighting that complex neural architectures are not always necessary for effective misinformation detection.
Gupta and Meel [2] investigated the effectiveness of the Passive-Aggressive Classifier (PAC) for real-time fake news detection. As an online learning algorithm, PAC remains passive when predictions are correct but aggressively updates its model parameters upon misclassification. This property makes it particularly suitable for dynamic environments where news data is continuously evolving.
Granik and Mesyura [3] analyzed the application of the Multinomial Naive Bayes classifier for detecting fake news. Despite its simplifying assumption of feature independence, the model was found to be highly efficient and effective, especially when applied to short news headlines. When combined with TF-IDF vectorization, Naive Bayes achieved competitive performance with minimal computational overhead.
Daniel et al. [4] conducted a study emphasizing the importance of feature representation in fake news detection. By employing Term Frequency–Inverse Document Frequency (TF-IDF) for transforming textual data into numerical form, they demonstrated that linear classifiers could effectively distinguish between misleading and factual content by leveraging term importance across the corpus.
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Traditional methods of detecting fake news, such as manual fact-checking and crowd-sourced reporting, are time-consuming, subjective, and difficult to scale. To overcome these limitations, AI-driven solutions are being developed that can automatically analyze and classify news content.

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